SRM TRP Engineering College
Tiruchirappalli
Course Assignment Instructions with Rubrics

Department	Artificial Intelligence and Data science										
Programme:	B. Tech			**Semester:**	V		**AY:**	2026 - 2027			
Faculty Handling:	Mrs. R. Kalaiyarasi										
Course Code & Title:	AD3501 - Deep Learning										
Assessment Task:				**Assessment Type:**	Term Work/Self				**Required Hrs:**		
Assessment Question based on:	*Open Ended*		✓	*Problem Based Learning*		✓	*Creative Thinking*				
	Close Ended			*Project Based Learning*			*Critical Thinking*			✓	
	Any Other Type		MCQ or any other	*Challenge Based Learning*		✓	*Level of Thinking (RBT)*			K5	
SDG to be Addressed:	SDG is Required		Yes/No	*SDG Number (Specify or Mention Students Choice)*			*Target Number (If Applicable)*				
SDG Integration Level: (If Addressed)	*Level 1: Awareness*			*Level 2: Design Alignment*		✓	*Level 3: Impact and Measurement*				
RIASEC Context	Realistic	Investigative ✓		Artistic	Social		Enterprising		Conventional ✓		
Presentation Format:	PowerPoint/Poster/Video/PDF/Written/Any other (Specify)										
Max Marks:	38										
Relevant Course Outcome(s):	10/9/26 All co's										
Submission Deadline:											

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Instructions to the Students:
Assessment Question: AI vision Guide
Guidelines: E.g. Guidelines for Preparation/Submission/Expected Output/Review Process/Presentation Format if any

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Rubrics for Evaluation:

Performance Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)

38/40

R. Kathy

AD3501-DEEP LEARNING

TEAM ASSIGNMENT-LITERATURE SURVEY

TOPIC: AI VISION GUIDE

AI-Driven Assistive Vision & Navigation Systems for the Visually Impaired (2025-2026 papers)

Guide/Subject Handling Faculty	Mrs.R.Kalaiyarasi,Assistant Professor,AI&DS
Department	Artificial Intelligence and Data Science 'C'
Institution	SRM TRP Engineering College,Trichy
Submission Date	On or before 03.08.2026

Team Members:

Yeshashwini.H - 814724243179

Samyuktha.R - 814724243127

	Paper Title & Year	Author(s)	Journal / Type	Methodology	Dataset(s)	Model	Result	Limitations
1	Visioneer: AI Guided Navigation for the Visually Impaired (2025)	Mohamed Areeb, Siddharth Kumar, Bhagyashree Patil	IJRASET (non-indexed)	Smart-glasses concept pairing a camera with YOLO object detection and landmark recognition; OpenCV handles real-time image processing and pyttsx3 converts recognised objects/cues into speech.	Not specified (system/prototype concept paper; no dataset reported).	YOLO object detector + OpenCV + pyttsx3 text-to-speech.	Describes a working smart-glasses design that detects obstacles and landmarks and delivers audio navigation cues; no quantitative accuracy figures are reported.	Concept/prototype stage only; no field trial or benchmark numbers given, so real-world reliability is unverified.
2	VISNAV: A Multimodal AI-Augmented Reality Navigation Aid for Visual Impairment Support (2025)	Krishi Sehrawat, M. Ahsani, Anagh C. Nambiar, Durgesh Kumar	IJISRT (non-indexed)	Combines computer vision with AR overlays and multimodal (audio, haptic, voice) feedback; uses YOLOv8/MobileNet-SSD for object recognition and AR-Kit/AR-Core for rendering navigation paths.	Not specified — evaluated in simulated environments rather than a named benchmark dataset.	YOLOv8 and MobileNet-SSD object detectors + AR-Kit/AR-Core.	Preliminary simulated-environment results suggest reduced reliance on external aids and an intuitive, cost-effective navigation experience; no numeric accuracy is reported.	Only tested in simulated settings so far; system design and methodology are presented but large-scale real-world validation is still pending.
3	Literature Review on AI-Based Real-Time Scene Understanding and Guidance for Visually Impaired Users (2025)	Sudha P, S. S	IEEE IConSCEPT 2025 (Scopus-indexed conference)	Literature review of a smartphone-based assistive pipeline combining on-device object recognition, traffic-signal awareness, OCR, spatial inference, text-to-speech and haptics, with an offline-first, privacy-preserving design focus.	Not applicable — this is a literature review, not an original experiment.	Not applicable (surveys multiple on-device CV/OCR/TTS model approaches across the literature).	Summarises model-selection trends, system workflow, and latency / real-time performance benchmarks reported across the surveyed assistive systems.	As a review, it validates no new system of its own; conclusions are only as reliable as the underlying studies it surveys.
4	AI-Enhanced Navigational Glasses for Blind (2025)	Shardul Mhetre, Pranav Naik, Soham Walimbe, Mrs. S. A. Bagal	IJFMR (non-indexed)	Survey of embedded platforms (Raspberry Pi, Arduino) paired with cameras and LiDAR for obstacle detection, alongside deep-learning techniques (CNNs, transformer vision models, SLAM) for perception, and multimodal audio/haptic interaction.	Not applicable — survey of the existing literature, no primary dataset used.	CNNs, transformer-based vision models, and SLAM, as reviewed across surveyed systems.	Identifies key challenges around latency, energy efficiency, cost, and user adaptability, and outlines emerging edge-AI / context-aware research directions.	Survey-level synthesis only; presents no original experimental validation or new prototype of its own.
5	AI Guide Dog: Egocentric Path Prediction on Smartphone (2025)	Aishwarya Jadhav, Jeffery Cao, Abhishree Shetty, Urvashi Priyam Kumar, Aditi Sharma, Ben Sukboontip, Jayant Sravan Tamarapalli, Jingyi Zhang, Anirudh Koul	arXiv preprint	Lightweight egocentric (first-person) navigation using a vision-only multi-label classifier to predict directional commands; integrates GPS and high-level directions for goal-based outdoor navigation and manages multi-path uncertainty for destination-free indoor navigation.	Custom AI Guide Dog (AIGD) dataset introduced by the authors, built for smartphone-based egocentric navigation.	Vision-only multi-label classification network, designed for real-time smartphone deployment.	Presented as the first system to handle both goal-oriented and exploratory navigation across indoor and outdoor settings, establishing a new benchmark for blind navigation assistance.	Preprint, not yet peer-reviewed; being a newly introduced benchmark, it has not yet been cross-validated by independent follow-up studies.
6	AI-Powered Assistive Navigation Systems for the Visually Impaired (2025)	M.V. Bramhe, P. Magare, Mayur Aglawe, Sayali Bamanpalliwar, Elasha Deoghare	Int. J. Adv. Comp. Electron. Commun. Tech. (non-indexed)	Review plus a proposed server-client architecture: a Raspberry Pi wearable with camera and ultrasonic sensor streams data to a local edge server that runs object detection, multisensor fusion, and optional large vision-language-model (LVLM) scene description with text-to-speech, backed by an on-device fallback detector.	Not applicable — review-based; the proposed architecture is not yet empirically tested by the authors.	Lightweight object detectors combined with large vision-language models (LVLMs).	Reports expected outcomes drawn from the surveyed literature: detection mAP in the ~75-90% range on optimised datasets, multi-fold latency reduction via server offloading, and fewer false positives through ultrasonic-vision fusion.	The proposed edge/server architecture itself is untested; network dependency, power constraints, LVLM compute cost, and privacy risk are flagged as open concerns.

S.No	Paper Title & Year	Author(s)	Journal / Type	Methodology	Dataset(s)	Model	Result	Limitations
7	Eye-Assist: Real-Time Visual Assistance for the Visually Impaired (2025)	Christopher Falcone, N. Brown, M. Salim, Zakaria Merzougui, Kruti Shah, Chetan Jaiswal	IEEE UEMCON 2025	Android application combining an on-device YOLOv8-TFLite model for detecting vehicles, pedestrians, stairs and traffic signals with an Intel RealSense D435i depth camera (RGB + depth + infrared) for spatial awareness, and Speech-to-Text/Text-to-Speech for hands-free interaction.	Not specified in the available summary; evaluated through on-device, real-time operation rather than a named benchmark set.	YOLOv8-TFLite on-device detector combined with RealSense RGB-D-IR depth sensing.	Reports improved situational awareness, mobility support, and better low-light performance from the added depth-camera data; no single overall accuracy figure is quoted.	Depends on specific hardware (Intel RealSense D435i), affecting portability and cost; no large-scale field-test numbers are reported.
8	Exploring Global Research Trends on AI-Driven Assistive Technologies for the Visually Impaired: A Bibliometric and Network Visualization Study (2025)	Fatima Abdul Habib Alyafei, Fareed Awae	Int. J. Acad. Res. Bus. Soc. Sci., HRMARS	Bibliometric and VOSviewer network-visualisation study of the global literature; a Scopus search covering 2015-2025 was screened using PRISMA 2020 procedures before descriptive and keyword-co-occurrence analysis.	403 Scopus records retrieved; 328 publications retained after PRISMA 2020 screening.	Not applicable — bibliometric/network-analysis study, not a technical scoring or vision model.	Shows a sharp rise in publications from 2020 onward, Computer Science/Engineering and India/US dominance, and core keyword clusters around assistive technology, AI, object detection and navigation.	Purely bibliometric; it maps research trends and gaps but does not evaluate the technical performance of any individual system.
9	Assistive Technologies for the Visually Impaired: A Review (2025)	Farzad Cheraghpour Samavati, Ali Rahimi Ghasem Abadi	Cureus Journal (peer-reviewed review)	Systematic review of 80 studies and commercial products (2000-2025) drawn from IEEE Xplore, PubMed, Scopus and Google Scholar, classified into visual-imagery, non-visual-data, map-based, 3D-sound, smartphone-based and commercial-product categories.	80 studies and commercial products reviewed across the 2000-2025 period.	Not applicable — review spanning many technology types rather than one model.	Highlights the growing role of deep learning, multimodal (vision+hearing+touch) integration and wearables, and calls for affordable, inclusive, user-centred design especially for low-resource settings.	Review-level synthesis; introduces no new experimental system or quantitative benchmark of its own.
10	Augmented Reality and Vision-Language Models to Guide Humans Across Manual Tasks (2025)	Luke Kottom, A. Stefan-Zavala, Morteza Gharib, Julian Humml	ACM conference paper	Mixed-reality assembly-guidance pipeline on an Apple Vision Pro AR headset: a CNN-based object detector tracks components while a vision-language model validates each assembly step against a state-space encoding and gives corrective textual/visual feedback.	Proof-of-concept case study: guiding users through building a LEGO set (no external benchmark dataset).	CNN-based foundational object detector + foundational vision-language model (VLM).	Users with no prior LEGO-assembly knowledge were guided step-by-step to a validated final build, with mistakes detected and corrective feedback given throughout.	Limited to a single proof-of-concept task (LEGO assembly); no formal quantitative accuracy metrics or larger user study are reported.
11	Guided Reality: Generating Visually-Enriched AR Task Guidance with LLMs and Vision Models (2025)	Ada Yi Zhao, Aditya Gunturu, E. Do, Ryo Suzuki	ACM conference paper	Fully automated AR system that uses an LLM to generate multi-step instructions from a user query, selects the appropriate visual-guidance type per step, extracts spatial interaction points with vision models, and embeds dynamic visual cues in physical space.	A corpus of user manuals for guidance generation, evaluated on real-world tasks via a user study.	LLM (instruction generation and guidance-type selection) + vision models (spatial-point extraction).	A user study (N=16) completing real-world tasks, plus qualitative feedback from four instructors, indicated the system supports task execution and could be integrated into training workflows.	Small user-study sample (16 participants); evaluation is largely qualitative rather than a large-scale quantitative benchmark.
12	Utilizing Vision-Language Models as Action Models for Intent Recognition and Assistance (2025)	Cesar Alan Contreras, Manolis Chiou, Alireza Rastegarpanah, Michal Szulik, Rustam Stolkin	arXiv preprint	Extends the authors' GUIDER framework by adding a vision-language model (VLM) and a text-only LLM as a semantic prior: YOLO detection and Segment-Anything-Model instance segmentation supply candidate object crops that the VLM scores for relevance to an operator's mission prompt.	Not yet reported — future evaluation is planned in Isaac Sim using a Franka Emika arm on a Ridgeback base.	YOLO + Segment Anything Model (SAM) + VLM/LLM semantic-prior scoring, layered onto the GUIDER navigation/manipulation framework.	Describes an architecture in which combined VLM/LLM belief scores trigger autonomy changes for navigation and manipulation once a threshold is exceeded; no experimental results are given yet.	No experimental results reported at time of writing — evaluation is explicitly future work, and it remains confined to simulation rather than real-world robots.
13	Guiding Medical Vision-Language Models with Diverse Visual Prompts:	Kangyu Zhu, Ziyuan Qin, Huahui Yi, Zekun Jiang,	NAACL 2025 (ACL Anthology, top-tier NLP venue)	Systematically studies diverse visual-prompt designs (e.g., markers, bounding boxes) as a way of steering	Medical-imaging benchmark data used to evaluate	Medical vision-language models tested under	Provides a comprehensive framework and comparison of how different visual-prompt designs affect medical	Focused on the medical-imaging domain rather than assistive navigation, so its findings transfer

S.No	Paper Title & Year	Author(s)	Journal / Type	Methodology	Dataset(s)	Model	Result	Limitations
	Framework Design and Comprehensive Exploration of Prompt Variations (2025)	Qicheng Lao, Shaoting Zhang, Kang Li		the behaviour of medical vision-language models, comparing prompt variations within a unified framework.	prompt variations (dataset specifics not detailed in the available abstract).	varying visual-prompt configurations.	VLM behaviour and guidance quality.	only indirectly to real-time assistive vision systems.
14	Vision-Language Models: A Review of Applications and Future Directions (2025)	Chenyu Guo	AETR conference proceedings (Madison Proceedings)	Reviews the three core technologies underpinning vision-language models: multimodal representation learning for a shared semantic space, cross-attention-based modal alignment, and hybrid fusion strategies for deep vision-text interaction.	Not applicable — review paper, no primary dataset.	Not applicable (surveys VLM architectures broadly rather than testing one model).	Outlines VLM applications in human-computer interaction, content creation, autonomous driving and medical health, and flags data-dependence, interpretability and compute-cost as open challenges.	General-purpose VLM review; not specific to assistive vision or navigation for the visually impaired.
15	Computer Vision Based Guidance, Navigation and Control for Autonomous Aerial Vehicles: A Systematic Survey (2026)	Cristian Ogneru, S. Raileanu	INCAS Bulletin (Scopus-indexed engineering journal)	PRISMA-guided systematic survey (2005-2025, six academic databases) mapping computer-vision techniques onto the five functional stages of autonomous flight (sensing, perception, navigation, guidance, control); compares VIO/vSLAM pipelines, one- and two-stage object detectors, and sensor-fusion approaches.	Literature corpus drawn from six academic databases under the PRISMA protocol (survey, not a primary experimental dataset).	ORB-SLAM3 (vSLAM), YOLOv8/YOLO-Tiny (one-stage detectors), R-CNN/Faster R-CNN (two-stage detectors).	Finds ORB-SLAM3 a robust backbone for GNSS-denied localisation, and one-stage detectors (YOLOv8/Tiny) exceeding 30 FPS on embedded GPUs, trading some accuracy against slower two-stage detectors.	Applies to autonomous aerial vehicles rather than human assistive navigation; notes limited robustness in extreme conditions and a lack of standard benchmarks/certification for AI perception.
16	Integrative Assistive Technologies for Visual Impairment: A Systematic Review of AI-Driven, Robotic, and Sensor-Fusion Systems (2026)	K. Premarathne, S. Rajapakshe, Pradeep Abeyagunawardena, C. Jayawardena	Journal of Robotics (Wiley/Hindawi, Scopus-indexed)	Systematic review (2017-2025) of assistive systems combining AI, robotics and sensor fusion for navigation, object detection, scene understanding and text reading for visually impaired users.	Literature corpus of devices/systems published 2017-2025 (systematic review, no primary dataset).	Multiple CV + robotics + multimodal-feedback systems surveyed collectively.	Finds that combining computer vision with robotics and multimodal feedback has produced hands-free, contactless alternatives to the white cane, improving precision, speed and usability.	Review-level only; flags persistent gaps in environment-specific object data, extensive outdoor field testing, and feedback methods beyond audio.
17	Improving Artificial Intelligence for Smart Guiding Stick Technology: A Real-Time Vision-Based Navigation Aid System (2026)	S. Balasubramanian, A. Vishnubharathi, M. Vidyaa Sri, K. Santhiya	IEEE ICDSAAI 2026	AI-driven smart stick using an upgraded CNN to interpret camera, inertial and ultrasonic sensor data for real-time scene interpretation; model compression and edge computing keep computation lightweight on embedded hardware.	Field testing with visually impaired users (sample size not specified in the available summary).	Upgraded CNN combined with multi-sensor (camera + inertial + ultrasonic) fusion, compressed for edge deployment.	Reported as more accurate, faster and more energy-efficient than typical smart sticks, with field-test users rating it easier to use, more reliable and more satisfying overall.	Specific quantitative accuracy/energy figures and field-test sample size are not detailed in the available abstract.
18	AI-Powered Conversational Navigation System for Visually Impaired Using Multi-Modal Real-time Sensing (2026)	A. Chavan, A. Sonawane, Swapnil Shinde, Aditya Shrirame	IJFMR (non-indexed)	Survey of 60+ peer-reviewed publications categorising navigation aids into electronic travel aids, position-locator devices, and electronic orientation aids, then proposes a 'Robo Guide' architecture combining high-resolution sensing, SLAM and conversational natural-language processing.	60+ peer-reviewed IEEE/ACM publications reviewed; no primary experimental dataset for the proposed Robo Guide system itself.	Proposed SLAM + conversational-NLP architecture ('Robo Guide') — conceptual, not yet implemented.	Identifies research gaps in adaptive learning and context-aware assistance, and proposes Robo Guide as a direction toward more personalised, socially acceptable navigation aids.	Robo Guide is a proposed architecture only, not yet built or evaluated; survey conclusions depend on the accuracy of the 60+ underlying reviewed papers.
19	NAVISIGHT: A Deep Learning and Voice-Assisted System for Intelligent Indoor Navigation of the Visually Impaired (2026)	Gurubunny T., H. M., Eekshitha K., Karapakula Balaji Sai Vrushadree, V. V	ISJEM (non-indexed)	Embedded AI system integrating YOLO-based object detection, semantic room classification, ultrasonic proximity sensing and IoT emergency alerting; synthesised voice prompts deliver standalone guidance	Not specified in the available summary; evaluated across diverse indoor environments.	YOLO object detector + semantic room classifier + ultrasonic sensor fusion.	Experimental evaluation reports high detection accuracy, low-latency performance and practical deployability across varied indoor settings (exact figures not given in the available abstract).	Standalone, no-external-processing design may limit scene understanding compared with cloud-augmented systems; precise quantitative metrics are not detailed in the summary.

S.No	Paper Title & Year	Author(s)	Journal / Type	Methodology	Dataset(s)	Model	Result	Limitations
				without external processing.				
20	AI-Assisted Vision Alarming System for Blind and Vision-Impaired People (2026)	Le Chung Tran, Sinh Khai Ly, Rhys Blacklidge, Jonathan Shemmell, Nathan Difford, Daniel Edward Cox, Theresa Harada	MDPI Sensors (Scopus-indexed)	Vision Alarming System (VAS) built on a Raspberry Pi 5 with a LiDAR sensor and an AI depth camera, run as separate Docker containers on a ROS 2 backbone; obstacle alerts are delivered over Bluetooth via the Piper neural text-to-speech engine and SoX audio processing.	Proof-of-concept device testing; no formal benchmark dataset specified.	AI depth-camera + LiDAR obstacle-detection pipeline on a ROS2-based architecture.	Proof-of-concept demonstrates a standalone, open-source, low-cost, low-power system that communicates obstacle position and nature to users via neural text-to-speech.	Still at proof-of-concept stage; no large-scale quantitative accuracy figures or user-trial results are reported yet.
21	A Systematic Review of Vision Language Models: Comprehensive Analysis of Architectures, Applications, Datasets and Challenges Towards Robust Multimodal Intelligence (2026)	Arifur Rahman	Array, Elsevier (Scopus-indexed)	PRISMA-guided systematic review (Jan 2021-Dec 2025) building a unified taxonomy of VLM architectures — dual-encoder, cross-attention fusion transformer, unified single-stream, and modular bridge (Q-Former/LoRA) models — and a multi-objective training framework combining contrastive, masked, and reinforcement-based alignment.	928 records screened across seven digital libraries; 48 articles retained for final synthesis, drawing on benchmarks such as ScienceQA, MIMIC-CXR, GQA and Winoground.	Taxonomy spanning CLIP-style dual-encoders, cross-attention fusion transformers, LLaVA-style unified models, and Q-Former/LoRA bridge models.	Ablations cited show a 31.7% accuracy drop on ScienceQA without contrastive pretraining (LLaVA-1.5) and an 18.2% drop in clinical-report accuracy on MIMIC-CXR without GRPO (MedVLM-R1); a formalised 'compositionality gap' explains 40-65% performance deficits on GQA/Winoground.	Most reviewed studies remain lab-based; the authors note a significant gap between benchmark performance and real-world, safety-critical deployment.

RESEARCH GAP IDENTIFICATION AND JUSTIFICATION

Identified Research Gap

Across the 21 papers surveyed, deep-learning and vision-language approaches to assistive navigation for the visually impaired show fast-growing technical sophistication — YOLOv8-class detectors, SLAM pipelines, AR overlays, and increasingly large vision-language models — but very few are validated with real visually impaired users under real-world conditions. Only Paper 17 (the smart guiding stick) explicitly reports field testing with visually impaired users; most others are prototype concepts, literature reviews, a bibliometric study, or systems evaluated only in simulation, with no quantitative accuracy figures against real users. The vision-language-model papers in this survey (Papers 10-14, 21) show VLMs succeeding in adjacent domains — assembly guidance, AR task instructions, medical imaging, general multimodal applications — but none apply and evaluate a VLM directly inside a real-time assistive-navigation pipeline for the visually impaired. Meanwhile the review papers (3, 4, 6, 9, 16) repeatedly flag the same unresolved constraints across the field: latency and compute limits on wearable hardware, dependency on network connectivity for heavier models, and a near-total lack of standardised outdoor, real-user field trials or shared benchmarks. There is therefore a clear gap for a system that pairs a lightweight always-on on-device detector with an edge/cloud-offloaded vision-language model for richer scene description, is explicitly evaluated through real-world field testing with visually impaired users rather than simulation alone, and reports quantitative accuracy and latency figures under an offline-first design.

Justification

- Validation gap: of the 21 papers, only one (Balasubramanian et al., Paper 17) explicitly reports field testing with visually impaired users; the majority are prototype/concept papers (Papers 1, 2, 7, 19, 20), reviews/surveys (Papers 3, 4, 6, 9, 14, 16), or a bibliometric study (Paper 8) with no primary user validation at all.
- Real-world vs. simulated conditions: VISNAV (Paper 2) is evaluated only in simulated environments, and the edge/server architecture proposed in Paper 6 is explicitly not yet tested by its own authors, leaving a gap between architecture design and outdoor, real-world reliability.
- VLM integration is unproven for assistive navigation specifically: Papers 10-14 and 21 demonstrate vision-language models in adjacent domains, but none of the surveyed navigation-specific systems (Papers 1, 2, 5, 7, 17, 19, 20) integrate or evaluate a VLM within their own real-time pipeline.
- Network dependency and compute cost recur as open constraints: Paper 6 proposes an edge/server split to manage this trade-off but does not test it, while Paper 4's survey and Paper 16's systematic review both flag latency, energy and connectivity as unresolved across the field.
- No shared benchmark or standard evaluation protocol exists across the reviewed assistive-navigation systems, unlike the general vision-language-model field surveyed in Paper 21, where common benchmarks (ScienceQA, MIMIC-CXR, GQA, Winoground) allow direct comparison across studies.

Proposed Direction (basis for the future Mini Project)

A promising direction for the mini-project is a hybrid assistive-navigation system that pairs a lightweight, always-on on-device detector (extending Papers 1, 5, 7, 19, 20) with an edge- or cloud-offloaded vision-language model for richer scene description (extending the untested architecture proposed in Paper 6), including an offline-first fallback mode for connectivity loss. The resulting system should be explicitly validated through field testing with visually impaired users across both indoor and outdoor settings — following the field-testing approach used in Paper 17 — with quantitative latency and accuracy figures reported, closing the validation gap that affects the majority of the papers in this survey.

SDG MAPPING

Mapped Sustainable Development Goals

The research gap identified above, and the proposed mini-project direction (a hybrid, field-validated, on-device plus edge-offloaded assistive navigation system), align with the following United Nations Sustainable Development Goals:

SDG No.	SDG Title	Relevance to the Project
SDG 3	Good Health and Well-Being	AI-driven assistive navigation directly supports the safety, independent mobility and well-being of the more than 2.2 billion people worldwide living with vision impairment (cited in Papers 6 and 7), reducing collision and injury risk during everyday navigation.
SDG 9	Industry, Innovation and Infrastructure	The range of techniques surveyed — embedded computer vision, large vision-language models, SLAM, AR/VR guidance and sensor fusion — reflects active innovation in applying AI infrastructure to assistive-technology hardware and software, which the proposed hybrid on-device/edge direction extends further.
SDG 10	Reduced Inequalities	Affordable, low-cost, open-source designs (Papers 1 and 20) and explicit calls for inclusive, low-resource-setting-friendly design (Paper 9) directly address unequal access to assistive technology, particularly for users in low-income regions where a cane or guide dog remains the only option.
SDG 11	Sustainable Cities and Communities	Real-time obstacle, traffic-signal and indoor/outdoor navigation support (Papers 1, 2, 3, 7, 17, 19 and 20) helps make urban streets and indoor spaces safely navigable and accessible for people with vision impairment, supporting inclusive, accessible cities.

Justification of Mapping

- The primary motivation across the papers reviewed is enabling safer, more independent mobility for visually impaired users, which links the work directly to SDG 3 (Good Health and Well-Being).
- The proposed hybrid direction — combining a lightweight on-device detector with an edge-offloaded vision-language model and an offline-first fallback — is itself an innovation-driven contribution to assistive-technology infrastructure, aligning with SDG 9 (Industry, Innovation and Infrastructure).
- Because commercial assistive devices remain costly or inaccessible in many regions, low-cost and open-source designs (as in Papers 1 and 20) support SDG 10 (Reduced Inequalities) by widening access to assistive navigation technology.
- Reliable, real-time obstacle and navigation support in both indoor and outdoor urban settings, as surveyed across multiple papers, connects the work to SDG 11 (Sustainable Cities and Communities) by making shared physical spaces more accessible.

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